

EXERCISES

Level I

(Problems based on Fundamentals)

DOT PRODUCT OF VECTORS

1. If $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 3\mathbf{j}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.
2. If $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, prove that $(\mathbf{a} + \mathbf{b})$ is perpendicular to $(\mathbf{a} - \mathbf{b})$.
3. If $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$, $|\mathbf{c}| = 5$ such that $(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{0}$, find the value of $(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$.
4. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are mutually perpendicular unit vectors, find $|\mathbf{a} + \mathbf{b} - \mathbf{c}|$.
5. Find the projection of the vector $\mathbf{a} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ on the vector $\mathbf{b} = \mathbf{j} - 2\mathbf{j} = \mathbf{i} - \mathbf{j}$.
6. Find the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ such that $2\mathbf{a} + \mathbf{b} = \mathbf{i} + \mathbf{j}$ and $\mathbf{a} + 2\mathbf{b} = \mathbf{i} - \mathbf{j}$.
7. Find the unit vector perpendicular to each of the vectors $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.
8. Find a unit vector in xy -plane that makes an angle of 45° with the vector $\mathbf{i} + \mathbf{j}$ and an angle of 60° with the vector $3\mathbf{i} - 4\mathbf{j}$.
9. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be mutually perpendicular vectors of equal magnitude, show that $(\mathbf{a} + \mathbf{b} + \mathbf{c})$ is equally inclined to $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.
10. Find a vector of magnitude $\sqrt{51}$ which makes equal angles with the vectors $\mathbf{a} = \frac{1}{3}(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$, $\mathbf{b} = \frac{1}{5}(-4\mathbf{i} - 3\mathbf{k})$ and $\mathbf{c} = \mathbf{j}$.
11. Find a vector of magnitude 4, which is equally inclined to vectors $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = \mathbf{k} + \mathbf{i}$.
12. Find a unit vector in the plane of $(\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ and $(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ and perpendicular to $(2\mathbf{i} + \mathbf{j} + \mathbf{k})$.
13. If $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are unit vectors and θ the angle between them, show that $\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}|\hat{\mathbf{a}} - \hat{\mathbf{b}}|$.
14. If $\hat{\mathbf{a}}$, and $\hat{\mathbf{b}}$ are unit vectors perpendicular to each other, find the value of $(\hat{\mathbf{a}} - \hat{\mathbf{b}}) \cdot (\hat{\mathbf{b}} - \hat{\mathbf{c}})$.
15. Let $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ be two unit vectors and θ the angle between them. If $(\hat{\mathbf{a}} + \hat{\mathbf{b}})$ is a unit vector, find $\sin(\theta)$.
16. If $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ be two unit vectors inclined to x -axis at angles 30° and 120° respectively, find the value of $|\hat{\mathbf{a}} + \hat{\mathbf{b}}|$.
17. If $\mathbf{a}$ and $\mathbf{b}$ are two vectors and their lengths are a and b , show that $\left(\frac{\mathbf{a}}{a^2} - \frac{\mathbf{b}}{b^2}\right)^2 = \left(\frac{\mathbf{a} - \mathbf{b}}{ab}\right)^2$.
18. Let $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} - \mathbf{j}$, and $\mathbf{w} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. If $\hat{\mathbf{n}}$ is a unit vector such that $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$ and $\mathbf{v} \cdot \hat{\mathbf{n}} = 0$, then find $|\mathbf{w} \cdot \hat{\mathbf{n}}|$.

19. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be unit vectors such that $\mathbf{a}$ is perpendicular to the plane of $\mathbf{b}$ and $\mathbf{c}$ and the angle between $\mathbf{b}$ and $\mathbf{c}$ is $\frac{\pi}{3}$, find $|\mathbf{a} + \mathbf{b} + \mathbf{c}|$.
20. If $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$, $|\mathbf{c}| = 5$ and $\mathbf{a} \perp (\mathbf{b} + \mathbf{c})$, $\mathbf{b} \perp (\mathbf{c} + \mathbf{a})$ and $\mathbf{c} \perp (\mathbf{a} + \mathbf{b})$, find $|\mathbf{a} + \mathbf{b} + \mathbf{c}|$.
21. Let $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are two unit vectors and the angle between them is 60° , find the value of $|\hat{\mathbf{a}} + \hat{\mathbf{b}}\hat{\mathbf{a}} - \hat{\mathbf{b}}|$.
22. If $|\mathbf{a}| = 1$, $|\mathbf{b}| = 4$, $|\mathbf{c}| = \frac{\sqrt{17}}{3}$ such that $(\mathbf{a} + \mathbf{b} + 3\mathbf{c}) = \mathbf{0}$, find the value of $(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$.
23. Constant forces $\mathbf{F}_1 = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{F}_2 = -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{F}_3 = \mathbf{j} - \mathbf{k}$ act on a particle at a point A. Find the work done when the particle is displaced from the point A to B where $\mathbf{A} = 4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 6\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.
24. Prove that, by vector method,

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}.$$
25. Prove that, by vector method,

$$a = b \cos(C) + c \cos(B).$$
26. Prove that, by vector method,

$$\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$$
27. Prove that, by vector method,

$$\cos(A + B) = \cos(A) \cos(B) - \sin(A) \sin(B)$$

CROSS PRODUCT OF TWO VECTORS

28. If $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, and $\mathbf{b} = \mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$, find $\mathbf{a} \times \mathbf{b}$.
29. Find the area of a triangle whose adjacent sides are determined by the vectors $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - 3\mathbf{j} + \mathbf{k}$.
30. Find the area of a parallelogram whose adjacent sides are determined by the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.
31. Find the area of a parallelogram whose diagonals are given by the vectors $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$.
32. Find the area of a triangle whose vertices are $(3, -1, 2)$, $(1, -1, -1)$ and $(4, -3, 1)$.
33. Find a unit vector perpendicular to each of the vectors $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $3\mathbf{i} - 4\mathbf{j} - \mathbf{k}$.
34. Prove that $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) + \mathbf{b} \times (\mathbf{c} + \mathbf{a}) + \mathbf{c} \times (\mathbf{a} + \mathbf{b}) = \mathbf{0}$.
35. If $\mathbf{a} \times \mathbf{b} = \mathbf{c} \times \mathbf{d}$ and $\mathbf{a} \times \mathbf{c} = \mathbf{b} \times \mathbf{d}$, show that $(\mathbf{a} - \mathbf{d})$ is parallel to $(\mathbf{b} - \mathbf{c})$.
36. If $(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{0}$, prove that $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$.

37. Prove that $|\mathbf{a} \times \hat{i}|^2 + |\mathbf{a} \times \hat{j}|^2 + |\mathbf{a} \times \hat{k}|^2 = 2a^2$.
38. Let $\hat{a}$, $\hat{b}$ and $\hat{c}$ be three unit vectors such that $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} = 0$ and the angle between $\mathbf{b}$ and $\mathbf{c}$ is $\frac{\pi}{6}$, prove that $\mathbf{a} = \pm 2(\mathbf{b} \times \mathbf{c})$.
39. If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, $\mathbf{a} \neq 0$ and $\mathbf{b} \neq \mathbf{c}$, prove that $\mathbf{b} = \mathbf{c} + \lambda \mathbf{a}$, $\forall \lambda \in R$.
40. Let $\mathbf{A} = 2\mathbf{i} + \mathbf{k}$, $\mathbf{B} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{C} = 4\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$. Find a vector $\mathbf{R}$ satisfying $\mathbf{R} \times \mathbf{B} = \mathbf{R} \times \mathbf{C}$ and $\mathbf{R} \cdot \mathbf{A} = 1$.
41. Find the number of vectors of unit length perpendicular to the vectors $\mathbf{a} = \mathbf{i} + \mathbf{j}$ and $\mathbf{b} = \mathbf{j} + \mathbf{k}$.
42. Let $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{b} = \mathbf{j}$, find a vector $\mathbf{c}$ for which $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ form a right handed system.
43. Let $\mathbf{a} = \mathbf{i} + \mathbf{j}$ and $\mathbf{b} = 2\mathbf{j} - \mathbf{k}$. Find the point of intersection of the lines $\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a}$ and $\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$.
44. If $\mathbf{a}$ and $\mathbf{b}$ are two non-zero vectors such that $\mathbf{r} \cdot \mathbf{a} = 0$ and $\mathbf{r} \times \mathbf{a} = \mathbf{b}$, find $\mathbf{r}$.
45. Let $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{c} = \mathbf{j} - \mathbf{k}$ such that $\mathbf{a} \cdot \mathbf{b} = 3$ and $\mathbf{a} \times \mathbf{b} = \mathbf{c}$, find $\mathbf{b}$.
46. Find the angle between the vectors $[\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}]$ and $(\mathbf{a} \times \mathbf{b})$.
47. Find the area of a square whose one diagonal is $(3\mathbf{i} + 4\mathbf{j})$.
48. A constant force $\mathbf{F} = 3\mathbf{i} + 2\mathbf{j} - 4$ is applied at the point $(1, -2, 2)$. Find the vector moment of $\mathbf{F}$ about the point $(2, -1, 3)$.
49. Forces $(5\mathbf{i} + \mathbf{k})$ and $(-5\mathbf{i} - \mathbf{k})$ act at the points $P(9, -1, 2)$ and $Q(3, -2, 1)$, respectively. Find the moment of the couple.
50. Find the torque about the point $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ of a force $4\mathbf{i} + \mathbf{k}$ acting through the point $(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$.
51. In a $\triangle ABC$, $a = BC$, $b = CA$ and $c = AB$, by using vector method, prove that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

52. Prove, by the vector method,

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

53. Prove, by vector method,

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

SCALAR TRIPLE PRODUCT OF VECTORS

54. For any three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, prove that

$$[\mathbf{a} + \mathbf{b}, \mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}] = 2[\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

55. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ be unit coplanar vectors, find

$$[2\mathbf{a} - \mathbf{b}, 2\mathbf{b} - \mathbf{c}, 2\mathbf{c} - \mathbf{a}].$$

56. Find $[\mathbf{a}, \mathbf{b} + \mathbf{c}, \mathbf{a} + \mathbf{b} + \mathbf{c}]$.

57. If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be three non-coplanar vectors, find $[\mathbf{a} + \mathbf{b} + \mathbf{c}, \mathbf{a} + \mathbf{b}, \mathbf{a} + \mathbf{c}]$.

58. Find the value of $[\mathbf{a} - \mathbf{b}, \mathbf{b} - \mathbf{c}, \mathbf{c} - \mathbf{a}]$.

59. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three non-zero and non-coplanar vectors such that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = |\mathbf{a}||\mathbf{b}||\mathbf{c}|$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

60. For non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 4$, find the value of $\frac{1}{2}(\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c})$.

61. If the vectors $2\mathbf{i} - \mathbf{j} + \lambda\mathbf{k}$, $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ are coplanar vectors, find the value of λ .

62. Find the volume of a parallelepiped whose coterminal edges are $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

63. Find the volume of a parallelepiped whose coterminal edges are $\mathbf{a} + \mathbf{b} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{b} + \mathbf{c} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\mathbf{c} + \mathbf{a} = 3\mathbf{i} + 4\mathbf{j} - 7\mathbf{k}$.

64. The volume of a parallelepiped determined by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ is 5. Find the volume of the parallelepiped determined by the vectors $3(\mathbf{a} + \mathbf{b})$, $(\mathbf{b} + \mathbf{c})$ and $2(\mathbf{c} + \mathbf{a})$.

65. If the volume of the parallelepiped formed by the vectors $\mathbf{i} + a\mathbf{j} + \mathbf{k}$, $\mathbf{j} + a\mathbf{k}$ and $a\mathbf{i} + \mathbf{k}$, $a > 0$, is minimum, find the value of a .

66. Let V_1 be the volume of the parallelepiped formed by three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ and V_2 be the volume of the parallelepiped formed by the vectors $\mathbf{p} = \mathbf{a} - 2\mathbf{b} + 2\mathbf{c}$, $\mathbf{q} = 3\mathbf{a} - \mathbf{b} + \mathbf{c}$ and $\mathbf{r} = \mathbf{a} - \mathbf{b} + 2\mathbf{c}$. Find $\left(\frac{V_1}{V_2}\right)$.

67. If the vectors $a\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{i} + b\mathbf{j} + \mathbf{k}$ and $\mathbf{i} + \mathbf{j} + c\mathbf{k}$, ($a, b, c \neq 1$) are coplanar, then find the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$.

68. For any three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, prove that

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}]^2 = \begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix}$$

69. The edges of a parallelepiped are of unit length and parallel to non-coplanar unit vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ such that $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = \frac{1}{2}$. Find the volume of the parallelepiped.

70. If $\mathbf{a}$ and $\mathbf{b}$ be non-zero and non-collinear vectors, prove that $[\mathbf{a}, \mathbf{b}, \mathbf{i}]\mathbf{i} + [\mathbf{a}, \mathbf{b}, \mathbf{j}]\mathbf{j} + [\mathbf{a}, \mathbf{b}, \mathbf{k}]\mathbf{k} = \mathbf{a} \times \mathbf{b}$.

71. If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{m}$ and $\mathbf{n}$ are non-zero vectors, prove that

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}](\mathbf{m} \times \mathbf{n}) = \begin{vmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \\ \mathbf{a} \cdot \mathbf{m} & \mathbf{b} \cdot \mathbf{m} & \mathbf{c} \cdot \mathbf{m} \\ \mathbf{a} \cdot \mathbf{n} & \mathbf{b} \cdot \mathbf{n} & \mathbf{c} \cdot \mathbf{n} \end{vmatrix}$$

72. If a vector $\mathbf{r}$ can be expressed as a linear combination of three non-coplanar vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, prove that

$$\mathbf{r} = \frac{[\mathbf{rbc}]}{[\mathbf{abc}]} \mathbf{a} + \frac{[\mathbf{rca}]}{[\mathbf{abc}]} \mathbf{b} + \frac{[\mathbf{rab}]}{[\mathbf{abc}]} \mathbf{c}$$

73. If $\mathbf{a}$ and $\mathbf{b}$ be unit vectors such that

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \frac{1}{4}, \text{ find the angle between } \mathbf{a} \text{ and } \mathbf{b}.$$

74. If $\alpha = x(\mathbf{a} + \mathbf{b}) + y(\mathbf{b} \times \mathbf{c}) + z(\mathbf{c} \times \mathbf{a})$ and $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \frac{1}{8}$, find $(x + y + z)$.

75. Find the volume of the tetrahedron with edges $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

76. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are unit vectors satisfying

$$|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 = 9, \text{ find } |2\mathbf{a} + 5\mathbf{b} + 5\mathbf{c}|.$$

77. The three vectors $\mathbf{i} + \mathbf{j}$, $\mathbf{j} + \mathbf{k}$, $\mathbf{k} + \mathbf{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to three planes form a parallelepiped.

Prove that its volume is $\frac{4}{\sqrt{3}}$ c.u.

78. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three non-coplanar vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \alpha \mathbf{d}$ and $\mathbf{b} + \mathbf{c} + \mathbf{d} = \beta \mathbf{a}$, find $\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} + 4$.

VECTOR TRIPLE PRODUCT

79. Find a unit vector which is orthogonal to $3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$ and is coplanar with the vectors $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{i} - \mathbf{j} + \mathbf{k}$.

80. If $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{a} \cdot \mathbf{b} = 1$, $\mathbf{a} \times \mathbf{b} = \mathbf{j} - \mathbf{k}$, find $\mathbf{b}$.

81. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three non-parallel unit vectors such that $(\mathbf{a} \times (\mathbf{b} \times \mathbf{c})) = \frac{1}{2}\mathbf{b}$, find the angles which $\mathbf{a}$ makes with $\mathbf{b}$ and $\mathbf{c}$.

82. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are three non-coplanar unit vectors such that

$$(\mathbf{a} \times (\mathbf{b} \times \mathbf{c})) = \frac{1}{\sqrt{2}}(\mathbf{b} + \mathbf{c}),$$

where $\mathbf{b}$ and $\mathbf{c}$ are non parallel, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

83. If $\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) + \mathbf{b} = 0$ such that $|\mathbf{a}| = 1$, $|\mathbf{b}|$ and $|\mathbf{c}| = 2$, find the angle between $\mathbf{a}$ and $\mathbf{c}$.

84. If $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is perpendicular to $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$, prove that $(\mathbf{a} \cdot \mathbf{c}) = 0$.

85. Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be three non-zero vectors such that

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \frac{1}{3}|\mathbf{b}||\mathbf{c}||\mathbf{a}|.$$

If θ be the angle between $\mathbf{b}$ and $\mathbf{c}$, find the value of $\sin \theta$.

86. If $\mathbf{a}$ and $\mathbf{b}$ be mutually perpendicular unit vectors satisfying $\mathbf{r} \cdot \mathbf{a} = 0$, $\mathbf{r} \cdot \mathbf{b} = 1$ and $[\mathbf{r}, \mathbf{a}, \mathbf{b}] = 1$, find $\mathbf{r}$.

87. If $|\mathbf{a}| = 7$, $|\mathbf{b}| = 1 = |\mathbf{c}|$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) = \frac{1}{2}\mathbf{a}$, where $\mathbf{a}$ and $\mathbf{b}$ are non-collinear vectors, find $|\mathbf{a} \times \mathbf{c}|$.

88. Simplify: $(\mathbf{d} + \mathbf{a}) \cdot \mathbf{a} \times \mathbf{b} \times (\mathbf{c} \times \mathbf{d})$.

89. Solve: $\mathbf{p} \times (\mathbf{x} \times \mathbf{a}) = \mathbf{b}$.

90. If $[\mathbf{x}, \mathbf{a}, \mathbf{b}] = 0$, $\mathbf{x} \cdot \mathbf{a} = 0$, $\mathbf{x} \cdot \mathbf{b} = 1$, $\mathbf{a} \cdot \mathbf{b} = 0$, find $\mathbf{x}$.

91. Find the value of

$$[\mathbf{a} \times (\mathbf{b} + \mathbf{c}), \mathbf{b} \times (\mathbf{c} - 2\mathbf{a}), \mathbf{c} \times (\mathbf{a} + 3\mathbf{b})]$$

if $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = 2$.

92. Find the value of

$$[(\mathbf{a} - \mathbf{b}), (\mathbf{a} - \mathbf{b} - \mathbf{c}), (\mathbf{a} + 2\mathbf{b} - \mathbf{c})] \text{ if } [\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

93. Find the value of

$$[(2\mathbf{a} \times 3\mathbf{b}), (3\mathbf{b} \times 4\mathbf{c}), (4\mathbf{c} \times 2\mathbf{a})] \text{ if } [\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

Vector Equations

94. Solve: $\mathbf{x} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-collinear vectors.

95. Solve: $\mathbf{x} \times \mathbf{a} = \mathbf{b}$ and $\mathbf{x} \cdot \mathbf{a} = 0$.

96. Solve: $\mathbf{x} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$ and $\mathbf{x} \cdot \mathbf{c} = 0$, where $\mathbf{c}$ is not perpendicular to $\mathbf{b}$.

97. Solve: $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = \mathbf{d}$.

98. Solve: $\mathbf{x} + \mathbf{y} = \mathbf{a}$, $\mathbf{x} \times \mathbf{y} = \mathbf{b}$, $\mathbf{x} \cdot \mathbf{a} = 1$

Hints & Solutions

Level - 1

1. Let θ the angle between a and b .

Now, $a \cdot b = ab \cos(\theta)$

$$\Rightarrow \cos(\theta) = \frac{a \cdot b}{ab}$$

$$\Rightarrow \cos(\theta) = \frac{6 - 6 + 0}{\sqrt{14}\sqrt{13}} = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

Hence, the angle between a and b is $\frac{\pi}{2}$.

2. We have,

$$\begin{aligned} a + b &= (i + 2j - 3k) + (3i + j + 2k) \\ &= (4i + 3j - k) \end{aligned}$$

Also,

$$\begin{aligned} a - b &= (i + 2j - 3k) - (3i + j + 2k) \\ &= (-2i + j - 5k) \end{aligned}$$

Now,

$$(a + b) \cdot (a - b) = -8 + 3 + 5 = 0$$

Thus, $(a + b)$ is perpendicular to $(a - b)$.

3. Given,

$$(a + b + c) = 0$$

$$\Rightarrow |(a + b + c)|^2 = 0$$

$$\Rightarrow |a|^2 + |b|^2 + |c|^2 + 2(a \cdot b + b \cdot c + c \cdot a) = 0$$

$$\Rightarrow 9 + 16 + 25 + 2(a \cdot b + b \cdot c + c \cdot a) = 0$$

$$\Rightarrow 2(a \cdot b + b \cdot c + c \cdot a) = -50$$

$$\Rightarrow (a \cdot b + b \cdot c + c \cdot a) = -25$$

4. We have,

$$\begin{aligned} & |(\mathbf{a} + \mathbf{b} - \mathbf{c})|^2 \\ &= |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}) \\ &= 1 + 1 + 1 + 0 \\ &= 3 \end{aligned}$$

$$\Rightarrow |\mathbf{a} + \mathbf{b} - \mathbf{c}| = \sqrt{3}$$

5. As we know that the projection of $\mathbf{a}$ on $\mathbf{b}$

$$\begin{aligned} &= \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} \\ &= \frac{3 - 2 + 2}{\sqrt{1 + 4 + 1}} = \frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}} \end{aligned}$$

6. Given $2\mathbf{a} + \mathbf{b} = \mathbf{i} + \mathbf{j}$ and $\mathbf{a} + 2\mathbf{b} = \mathbf{i} - \mathbf{j}$

Solving, we get

$$\mathbf{a} = \frac{1}{3}(\mathbf{i} + 3\mathbf{j}) \text{ and } \mathbf{b} = \frac{1}{3}(\mathbf{i} - 3\mathbf{j})$$

Since, $\mathbf{a} \cdot \mathbf{b} = ab \cos(\theta)$

$$\Rightarrow \cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{ab}$$

$$\Rightarrow \cos(\theta) = \frac{\frac{1}{9}(1 - 9)}{\frac{10}{9}} = -\frac{8}{10} = -\frac{4}{5}$$

$$\Rightarrow (\theta) = \cos^{-1}\left(-\frac{4}{5}\right)$$

7. Let $\mathbf{r} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$.

We have to find $\hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|}$.

Here, $\mathbf{r}$ is perpendicular to the given vectors, so

$$a - 2b + c = 0$$

$$\text{and } 2a + b - 3c = 0$$

Solving, we get

$$\frac{a}{6-1} = \frac{b}{2+3} = \frac{c}{1+4}$$

$$\Rightarrow \frac{a}{5} = \frac{b}{5} = \frac{c}{5}$$

$$\Rightarrow a = b = c = \lambda (\text{say})$$

$$\text{Thus, } \hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{\lambda(\mathbf{i} + \mathbf{j} + \mathbf{k})}{\lambda\sqrt{3}} = \frac{(\mathbf{i} + \mathbf{j} + \mathbf{k})}{\sqrt{3}}$$

8. Let the vector be $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$.

We have to find $\hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|}$.

$$\text{Let } \mathbf{a} = \mathbf{i} + \mathbf{j} \text{ and } \mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$$

Now,

$$\mathbf{r} \cdot \mathbf{a} = ra \cos(45^\circ)$$

$$\Rightarrow x + y = \sqrt{x^2 + y^2} \cdot \sqrt{2} \cdot \frac{1}{\sqrt{2}}$$

$$\Rightarrow x + y = \sqrt{x^2 + y^2} \quad \dots(i)$$

and

$$\mathbf{r} \cdot \mathbf{b} = rb \cos(60^\circ)$$

$$\Rightarrow 3x - 4y = \sqrt{x^2 + y^2} \cdot 5 \cdot \frac{1}{2}$$

$$\Rightarrow \frac{2}{5}(3x - 4y) = \sqrt{x^2 + y^2} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$x = 13y$$

$$\text{Thus, } \hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{x\mathbf{i} + y\mathbf{j}}{\sqrt{x^2 + y^2}}$$

$$= \frac{13y\mathbf{i} + y\mathbf{j}}{\sqrt{169y^2 + y^2}}$$

$$= \frac{13\mathbf{i} + \mathbf{j}}{\sqrt{170}}$$

9. Given $\mathbf{a} \cdot \mathbf{b} = 0 = \mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{c}$

$$\text{Let } |\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = \lambda$$

$$\text{Now, } (\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot \mathbf{a} = |\mathbf{a} + \mathbf{b} + \mathbf{c}| |\mathbf{a}| \cos(\theta_1)$$

$$\Rightarrow |\mathbf{a} + \mathbf{b} + \mathbf{c}| |\mathbf{a}| \cos(\theta_1) = (\mathbf{a} \cdot \mathbf{a} + \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c})$$

$$\Rightarrow |\mathbf{a} + \mathbf{b} + \mathbf{c}| |\mathbf{a}| \cos(\theta_1) = |\mathbf{a}|^2$$

$$\Rightarrow \cos(\theta_1) = \frac{|\mathbf{a}|}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|} = \frac{\lambda}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|}$$

$$\text{Similarly, } \cos(\theta_2) = \frac{|\mathbf{b}|}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|} = \frac{\lambda}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|}$$

$$\text{and } \cos(\theta_3) = \frac{|\mathbf{c}|}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|} = \frac{\lambda}{|\mathbf{a} + \mathbf{b} + \mathbf{c}|}$$

$$\text{Thus, } \cos(\theta_1) = \cos(\theta_2) = \cos(\theta_3)$$

$$\Rightarrow (\theta_1) = (\theta_2) = (\theta_3).$$

10. Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ such that $|\mathbf{r}| = \sqrt{51}$.

It is given that

$$\cos(\theta_1) = \cos(\theta_2) = \cos(\theta_3)$$

$$\Rightarrow \frac{\mathbf{a} \cdot \mathbf{r}}{|\mathbf{a}| |\mathbf{r}|} = \frac{\mathbf{b} \cdot \mathbf{r}}{|\mathbf{b}| |\mathbf{r}|} = \frac{\mathbf{c} \cdot \mathbf{r}}{|\mathbf{c}| |\mathbf{r}|}$$

$$\Rightarrow \frac{\mathbf{a} \cdot \mathbf{r}}{|\mathbf{a}|} = \frac{\mathbf{b} \cdot \mathbf{r}}{|\mathbf{b}|} = \frac{\mathbf{c} \cdot \mathbf{r}}{|\mathbf{c}|}$$

$$\Rightarrow \mathbf{a} \cdot \mathbf{r} = \mathbf{b} \cdot \mathbf{r} = \mathbf{c} \cdot \mathbf{r}$$

$$(\because |\mathbf{a}| = 1 = |\mathbf{b}| = |\mathbf{c}|)$$

$$\Rightarrow \frac{x - 2y + 2z}{3} = \frac{-4x - 3z}{5} = \frac{y}{1}$$

$$\Rightarrow \frac{x - 2y + 2z}{3} = \frac{-4x - 3z}{5} = \frac{y}{1} = \lambda(\text{say})$$

Solving, we get

$$x = -5\lambda, y = \lambda, z = 5\lambda$$

Also,

$$|\mathbf{r}| = \sqrt{51}$$

$$\Rightarrow x^2 + y^2 + z^2 = 51$$

$$\Rightarrow 25\lambda^2 + \lambda^2 + 25\lambda^2 = 51$$

$$\Rightarrow 51\lambda^2 = 51$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

$$\Rightarrow \mathbf{r} = \pm(-5\mathbf{i} + \mathbf{j} + 5\mathbf{k})$$

11. Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ such that $|\mathbf{r}| = 4$.

It is given that

$$\cos(\theta_1) = \cos(\theta_2) = \cos(\theta_3)$$

$$\Rightarrow \frac{\mathbf{a} \cdot \mathbf{r}}{|\mathbf{a}||\mathbf{r}|} = \frac{\mathbf{b} \cdot \mathbf{r}}{|\mathbf{b}||\mathbf{r}|} = \frac{\mathbf{c} \cdot \mathbf{r}}{|\mathbf{c}||\mathbf{r}|}$$

$$\Rightarrow \frac{\mathbf{a} \cdot \mathbf{r}}{|\mathbf{a}|} = \frac{\mathbf{b} \cdot \mathbf{r}}{|\mathbf{b}|} = \frac{\mathbf{c} \cdot \mathbf{r}}{|\mathbf{c}|}$$

$$\Rightarrow \mathbf{a} \cdot \mathbf{r} = \mathbf{b} \cdot \mathbf{r} = \mathbf{c} \cdot \mathbf{r}$$

$$\left(\because |\mathbf{a}| = \frac{1}{\sqrt{2}} = |\mathbf{b}| = |\mathbf{c}| \right)$$

$$\Rightarrow x + y = y + z = x + z$$

$$\Rightarrow x + y = y + z = x + z = \lambda(\text{say})$$

Solving, we get

$$x = -\frac{1}{2} = y = z$$

Also,

$$|\mathbf{r}| = 4$$

$$\Rightarrow x^2 + y^2 + z^2 = 4$$

$$\Rightarrow \frac{\lambda^2}{4} + \frac{\lambda^2}{4} + \frac{\lambda^2}{4} = 4$$

$$\Rightarrow \frac{3\lambda^2}{4} = 4$$

$$\Rightarrow \lambda^2 = \frac{16}{3}$$

$$\Rightarrow \lambda = \pm \frac{4}{\sqrt{3}}$$

$$\text{Therefore, } \mathbf{r} = \pm \frac{4}{\sqrt{3}} (\mathbf{i} + \mathbf{j} + \mathbf{k})$$

$$\mathbf{b} = (\mathbf{i} + \mathbf{j} + 2\mathbf{k})$$

$$\text{and } \mathbf{c} = (2\mathbf{i} + \mathbf{j} + \mathbf{k})$$

and $\mathbf{r}$ be a vector

such that $\mathbf{r} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$.

It is given that $\mathbf{r}$, $\mathbf{a}$ and $\mathbf{b}$ lie in the same plane, so

$$\mathbf{r} \cdot (\mathbf{a} \times \mathbf{b}) = 0$$

$$\Rightarrow \begin{vmatrix} x & y & z \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 3x - y - z = 0 \quad \dots(i)$$

Also,

$$\mathbf{r} \cdot \mathbf{c} = 0$$

$$\Rightarrow 2x + y + z = 0 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$\frac{x}{0} = \frac{y}{-5} = \frac{z}{5}$$

$$\Rightarrow \frac{x}{0} = \frac{y}{-1} = \frac{z}{1} = \lambda(\text{say})$$

$$\text{Therefore, } \hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|}$$

$$\Rightarrow \hat{\mathbf{r}} = \pm(\mathbf{j} + \mathbf{k})$$

13. We have,

$$\begin{aligned} & |(\hat{\mathbf{a}} - \hat{\mathbf{b}})|^2 \\ &= |\hat{\mathbf{a}}|^2 + |\hat{\mathbf{b}}|^2 - 2(\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}) \\ &= [1 + 1 - 2(\hat{\mathbf{a}} \cdot \hat{\mathbf{b}})] \\ &= 2 - 2(\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}) \\ &= 2 - 2\cos\theta \\ &= 2(1 - \cos\theta) \end{aligned}$$

$$= 2 \times 2\sin^2\left(\frac{\theta}{2}\right)$$

$$\Rightarrow |\hat{\mathbf{a}} - \hat{\mathbf{b}}| = 2\sin\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \sin\left(\frac{\theta}{2}\right) = \frac{1}{2} |\hat{\mathbf{a}} - \hat{\mathbf{b}}|$$

14. Given $\hat{\mathbf{a}} \cdot \hat{\mathbf{b}} = 0 = \hat{\mathbf{b}} \cdot \hat{\mathbf{c}} = \hat{\mathbf{c}} \cdot \hat{\mathbf{a}}$

$$\text{and } |\hat{\mathbf{a}}| = 1 = |\hat{\mathbf{b}}| = |\hat{\mathbf{c}}|$$

We have,

$$\begin{aligned} & (\hat{\mathbf{a}} - \hat{\mathbf{b}}) \cdot (\hat{\mathbf{b}} - \hat{\mathbf{c}}) \\ &= (\hat{\mathbf{a}} \cdot \hat{\mathbf{b}} - \hat{\mathbf{a}} \cdot \hat{\mathbf{c}} - \hat{\mathbf{b}} \cdot \hat{\mathbf{b}} + \hat{\mathbf{b}} \cdot \hat{\mathbf{c}}) \\ &= (0 - 0 - \hat{\mathbf{b}} \cdot \hat{\mathbf{b}} + 0) \end{aligned}$$

12. Let $\mathbf{a} = (\mathbf{i} + 2\mathbf{j} + \mathbf{k})$,

$$= -|\hat{b}|^2$$

$$= -1.$$

15. Given $|\hat{a}| = 1 = |\hat{b}| = |\hat{a} + \hat{b}|$

We have,

$$|\hat{a} + \hat{b}| = 1$$

$$\Rightarrow |\hat{a} + \hat{b}|^2 = 1$$

$$\Rightarrow |\hat{a}|^2 + |\hat{b}|^2 + 2(\hat{a} \cdot \hat{b}) = 1$$

$$\Rightarrow 1 + 1 + 2(\hat{a} \cdot \hat{b}) = 1$$

$$\Rightarrow 2\cos(\theta) = 1 - 2 = -1$$

$$\Rightarrow \cos(\theta) = -\frac{1}{2}$$

Thus, $\sin(\theta) = \sqrt{1 - \cos^2(\theta)}$

$$\Rightarrow \sin(\theta) = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$$

16. It is given that, $\mathbf{a} \cdot \hat{\mathbf{i}} = |\mathbf{a}||\hat{\mathbf{i}}| \cos(30^\circ) = \frac{\sqrt{3}}{2}$

and $\mathbf{b} \cdot \hat{\mathbf{i}} = |\mathbf{b}||\hat{\mathbf{i}}| \cos(120^\circ) = -\frac{1}{2}$

Clearly, the angle between $\mathbf{a}$ and $\mathbf{b}$ is 90° .

Thus, $(\hat{a} \cdot \hat{b}) = 0$

We have,

$$|\hat{a} + \hat{b}|^2 = |\hat{a}|^2 + |\hat{b}|^2 + 2(\hat{a} \cdot \hat{b})$$

$$= 1 + 1 + 0$$

$$= 2$$

$$\Rightarrow |\hat{a} + \hat{b}| = \sqrt{2}$$

17. Given,

$$\left(\frac{\mathbf{a}}{a^2} - \frac{\mathbf{b}}{b^2}\right)^2 = \frac{a^2}{a^4} + \frac{b^2}{b^4} - \frac{2(\mathbf{a} \cdot \mathbf{b})}{a^2 b^2}$$

$$= \frac{1}{a^2} + \frac{1}{b^2} - \frac{2(\mathbf{a} \cdot \mathbf{b})}{a^2 b^2}$$

$$= \frac{a^2 + b^2 - 2(\mathbf{a} \cdot \mathbf{b})}{a^2 b^2}$$

$$= \left(\frac{\mathbf{a} - \mathbf{b}}{ab}\right)^2$$

18. Let $\hat{\mathbf{n}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ such that $|\hat{\mathbf{n}}| = 1$.

Now,

$$\mathbf{u} \cdot \hat{\mathbf{n}} = 0$$

$$\Rightarrow x + y = 0$$

$$\Rightarrow x = -y$$

...(i)

Also, $\mathbf{v} \cdot \hat{\mathbf{n}} = 0$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x = y$$

...(ii)

From Eqs. (i) and (ii), we get

$$x = 0 = y$$

Also,

$$|\hat{\mathbf{n}}| = 1$$

$$\Rightarrow \sqrt{x^2 + y^2 + z^2} = 1$$

$$\Rightarrow x^2 + y^2 + z^2 = 1$$

$$\Rightarrow 0 + 0 + z^2 = 1$$

$$\Rightarrow z = \pm 1$$

Thus, $\mathbf{r} = \pm \mathbf{k}$

Now, $|(\mathbf{w} \cdot \hat{\mathbf{n}})| = |-3| = 3$.

19. Given $\mathbf{a} \cdot \mathbf{b} = 0 = \mathbf{a} \cdot \mathbf{c}$

Also, $\mathbf{b} \cdot \mathbf{c} = |\mathbf{b}||\mathbf{c}| \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

Now,

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2$$

$$= |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$$

$$= 1 + 1 + 1 + 2\left(0 + \frac{1}{2} + 0\right)$$

$$= 4$$

Thus, $|\mathbf{a} + \mathbf{b} + \mathbf{c}| = 2$.

20. Given

$$2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 0$$

Now,

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2$$

$$= |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$$

$$= 9 + 16 + 25 + 0$$

$$= 50$$

Thus, $2|\mathbf{a} + \mathbf{b} + \mathbf{c}| = 5$.

21. We have,

$$\frac{|\hat{a} + \hat{b}|^2}{|\hat{a} - \hat{b}|^2} = \frac{|\hat{a} + \hat{b}|^2}{|\hat{a} - \hat{b}|^2}$$

$$= \frac{|\hat{a}|^2 + |\hat{b}|^2 + 2(\hat{a} \cdot \hat{b})}{|\hat{a}|^2 + |\hat{b}|^2 - 2(\hat{a} \cdot \hat{b})}$$

$$= \frac{1 + 1 + 2\cos(60^\circ)}{1 + 1 - 2\cos(60^\circ)}$$

$$= \frac{1 + 1 + 1}{1 + 1 - 1}$$

$$= 3.$$

Thus, $\left| \frac{\hat{a} + \hat{b}}{\hat{a} - \hat{b}} \right| = \sqrt{3}$

22. Given,

$$(\mathbf{a} + \mathbf{b} + 3\mathbf{c}) = \mathbf{0}$$

$$\Rightarrow (\mathbf{a} + \mathbf{b} + \mathbf{c}) = (-2\mathbf{c})$$

We have,

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = |(-2\mathbf{c})|^2$$

$$\Rightarrow |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 4|\mathbf{c}|^2$$

$$\Rightarrow 1 + 16 + \frac{17}{9} + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = \frac{68}{9}$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = \frac{68 - 17}{9} - 17$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = \frac{51}{9} - 17$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = \frac{17}{3} - 17$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = -\frac{34}{3}$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = -\frac{17}{3}$$

23. Let the resultant force be $\mathbf{F}$ such that

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$$

i.e. $\mathbf{F} = 2\mathbf{j} - \mathbf{k}$

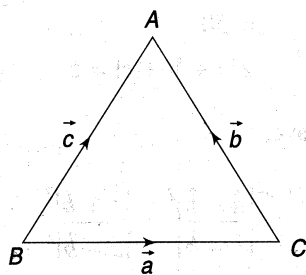
Now, $\mathbf{AB} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$

$$\therefore \text{Work done} = (\text{Force}) \cdot (\text{Displacement})$$

$$= (2\mathbf{j} - \mathbf{k}) \cdot (2\mathbf{i} + 4\mathbf{j} - \mathbf{k})$$

$$= 8 + 1 = 9.$$

24. Let $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$ and $\overrightarrow{AC} = \vec{c}$



We have $\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = \mathbf{0}$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = \mathbf{0}$$

$$\Rightarrow \vec{b} + \vec{c} = -\vec{a}$$

$$\Rightarrow (\vec{b} + \vec{c})^2 = (-\vec{a})^2$$

$$\Rightarrow b^2 + c^2 + 2b \cdot c = a^2$$

$$\Rightarrow b^2 + c^2 + 2bcc \cos(\pi - A) = a^2$$

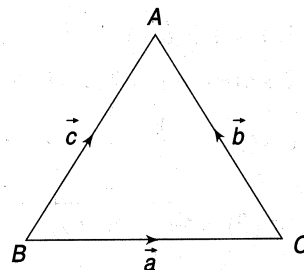
$$\Rightarrow b^2 + c^2 + 2bcc \cos(A) = a^2$$

$$\Rightarrow 2bcc \cos(A) = b^2 + c^2 - a^2$$

$$\Rightarrow \cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$

Hence, the result.

25. Let $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$ and $\overrightarrow{AB} = \vec{c}$



We have,

$$\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = \vec{0}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{b} + \vec{c} = -\vec{a}$$

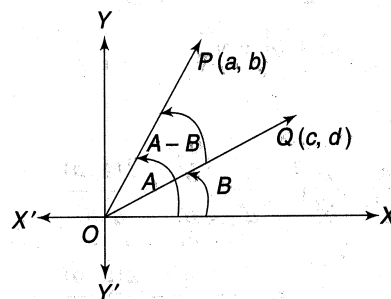
$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = -\vec{a} \cdot \vec{a}$$

$$\Rightarrow ab \cos(\pi - C) + ac \cos(\pi - B) = -a^2$$

$$\Rightarrow -ab \cos(C) - ac \cos(B) = -a^2$$

$$\Rightarrow a = b \cos(C) + c \cos(B)$$

26.



Here, $\overrightarrow{OP} = a\hat{i} + b\hat{j}$

and $\overrightarrow{OQ} = c\hat{i} + d\hat{j}$

Now, $\overrightarrow{OP} \cdot \overrightarrow{OQ} = |\overrightarrow{OP}| |\overrightarrow{OQ}| \cos(\angle POQ)$

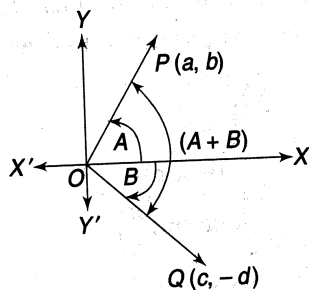
Now, $(a\hat{i} + b\hat{j}) \cdot (c\hat{i} + d\hat{j})$

$$= \sqrt{a^2 + b^2} \sqrt{c^2 + d^2} \cos(A - B)$$

$$\Rightarrow \cos(A - B)$$

$$\begin{aligned}
 &= \frac{(a\hat{i} + b\hat{j}) \cdot (c\hat{i} + d\hat{j})}{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}} \\
 &= \frac{ac + bd}{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}} \\
 &= \frac{a}{\sqrt{a^2 + b^2}} \cdot \frac{c}{\sqrt{c^2 + d^2}} + \frac{b}{\sqrt{a^2 + b^2}} \cdot \frac{d}{\sqrt{c^2 + d^2}} \\
 &= \cos A \cdot \cos B + \sin A \cdot \sin B
 \end{aligned}$$

27.



Here, $\vec{OP} = a\hat{i} + b\hat{j}$

and $\vec{OQ} = c\hat{i} + d\hat{j}$

Now,

$$\begin{aligned}
 \vec{OP} \cdot \vec{OQ} &= |\vec{OP}| |\vec{OQ}| \cos(\angle POQ) \\
 \Rightarrow ac - bd &= (OP)(OQ) \cos(A + B) \\
 \Rightarrow \cos(A + B) &= \frac{ac - bd}{(OP)(OQ)} \\
 &= \frac{a}{OP} \cdot \frac{c}{OQ} - \frac{b}{OP} \cdot \frac{d}{OQ} \\
 &= \cos(A) \cos(B) - \sin(A) \sin(B)
 \end{aligned}$$

28. We have,

$$\begin{aligned}
 \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 1 & -3 & -5 \end{vmatrix} \\
 &= 8\mathbf{i} + 11\mathbf{j} - 5\mathbf{k}
 \end{aligned}$$

29. Area of a triangle = $\frac{1}{2} |\mathbf{a} \times \mathbf{b}|$

Now,

$$\begin{aligned}
 \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 0 \\ 5 & 7 & 0 \end{vmatrix} \\
 &= (21 - 20)\mathbf{k} \\
 &= \mathbf{k}
 \end{aligned}$$

Thus, the area of the triangle = $\frac{1}{2}$ sq.u

30. Now,

$$\begin{aligned}
 \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix} \\
 &= 8\mathbf{i} + 8\mathbf{j} - 8\mathbf{k} \\
 &= 8(\mathbf{i} + \mathbf{j} - \mathbf{k})
 \end{aligned}$$

Area of the parallelogram

$$\begin{aligned}
 &= |\mathbf{a} \times \mathbf{b}| \\
 &= 8\sqrt{3} \text{ sq.u.}
 \end{aligned}$$

31. Now,

$$\begin{aligned}
 \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 3 & -4 & 1 \end{vmatrix} \\
 &= 5\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}
 \end{aligned}$$

Area of the parallelogram

$$\begin{aligned}
 &= \frac{1}{2} |\mathbf{a} \times \mathbf{b}| \\
 &= \frac{1}{2} \times \sqrt{25 + 4 + 49} \\
 &= \frac{\sqrt{78}}{2} \text{ sq.u.}
 \end{aligned}$$

32. Let

$$\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k},$$

$$\mathbf{b} = \mathbf{i} - \mathbf{j} - \mathbf{k}$$

and

$$\mathbf{c} = 4\mathbf{i} - 3\mathbf{j} + \mathbf{k}.$$

Now,

$$\mathbf{ab} = -2\mathbf{i} - 3\mathbf{k}$$

and

$$\mathbf{ac} = \mathbf{i} - 2\mathbf{j} - \mathbf{k}$$

$$\therefore \mathbf{ab} \times \mathbf{ac} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 0 & -3 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= -6\mathbf{i} - 5\mathbf{j} + 4\mathbf{k}$$

Area of the triangle = $\frac{1}{2} |\mathbf{ab} \times \mathbf{ac}|$

$$= \frac{1}{2} \sqrt{36 + 25 + 16} = \frac{\sqrt{77}}{2} \text{ sq.u}$$

33. Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j} - \mathbf{k}$.

A unit vector perpendicular to $\mathbf{a}$ and $\mathbf{b}$

$$= \pm \frac{(\mathbf{a} \times \mathbf{b})}{|\mathbf{a} \times \mathbf{b}|}$$

Now,

$$\begin{aligned}
 (\mathbf{a} \times \mathbf{b}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 3 & -4 & -1 \end{vmatrix} \\
 &= 5\mathbf{i} + 5\mathbf{j} - 5\mathbf{k} \\
 &= 5(\mathbf{i} + \mathbf{j} - \mathbf{k})
 \end{aligned}$$

Thus, the unit vector = $\pm \frac{(\mathbf{a} \times \mathbf{b})}{|\mathbf{a} \times \mathbf{b}|}$

$$= \pm \frac{1}{\sqrt{3}} (\mathbf{i} + \mathbf{j} - \mathbf{k}).$$

34. We have,

$$\begin{aligned} & \mathbf{a} \times (\mathbf{b} + \mathbf{c}) + \mathbf{b} \times (\mathbf{c} + \mathbf{a}) + \mathbf{c} \times (\mathbf{a} + \mathbf{b}) \\ &= \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a} + \mathbf{c} \times \mathbf{b} \\ &= \mathbf{a} \times \mathbf{b} - \mathbf{c} \times \mathbf{a} + \mathbf{b} \times \mathbf{c} - \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a} - \mathbf{b} \times \mathbf{c} \\ &= \mathbf{0} \end{aligned}$$

35. Given $\mathbf{a} \times \mathbf{b} = \mathbf{c} \times \mathbf{d}$... (i)

and $\mathbf{a} \times \mathbf{c} = \mathbf{b} \times \mathbf{d}$... (ii)

Subtracting Eq. (ii) from Eq. (i), we get

$$\begin{aligned} & \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} = \mathbf{c} \times \mathbf{d} - \mathbf{b} \times \mathbf{d} \\ \Rightarrow & \mathbf{a} \times (\mathbf{b} - \mathbf{c}) = (\mathbf{c} - \mathbf{b}) \times \mathbf{d} \\ \Rightarrow & \mathbf{a} \times (\mathbf{b} - \mathbf{c}) - (\mathbf{c} - \mathbf{b}) \times \mathbf{d} = \mathbf{0} \\ \Rightarrow & \mathbf{a} \times (\mathbf{b} - \mathbf{c}) + (\mathbf{c} - \mathbf{b}) \times \mathbf{d} = \mathbf{0} \\ \Rightarrow & \mathbf{a} \times (\mathbf{b} - \mathbf{c}) - \mathbf{d} \times (\mathbf{b} - \mathbf{c}) = \mathbf{0} \\ \Rightarrow & (\mathbf{a} - \mathbf{d}) \times (\mathbf{b} - \mathbf{c}) = \mathbf{0} \end{aligned}$$

Thus, $(\mathbf{a} - \mathbf{d})$ is parallel to $(\mathbf{b} - \mathbf{c})$.

36. Given,

$$\begin{aligned} & (\mathbf{a} + \mathbf{b} + \mathbf{c}) \\ \Rightarrow & \mathbf{a} \times (\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{0} = \mathbf{0} \\ \Rightarrow & \mathbf{a} \times \mathbf{a} + \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} = \mathbf{0} \\ \Rightarrow & \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} = \mathbf{0} \\ \Rightarrow & \mathbf{a} \times \mathbf{b} = -\mathbf{a} \times \mathbf{c} \\ \Rightarrow & \mathbf{a} \times \mathbf{b} = \mathbf{c} \times \mathbf{a} \end{aligned} \quad \dots (i)$$

Also,

$$\begin{aligned} & (\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{0} \\ \Rightarrow & \mathbf{b} \times (\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{b} \times \mathbf{0} = \mathbf{0} \\ \Rightarrow & \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} = \mathbf{0} \\ \Rightarrow & \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c} = \mathbf{0} \\ \Rightarrow & \mathbf{b} \times \mathbf{c} = -\mathbf{b} \times \mathbf{a} \\ \Rightarrow & \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b} \end{aligned} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$$

37. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Now, $(\mathbf{a} \times \mathbf{i})$

$$\begin{aligned} &= (a_1\mathbf{i} \times \mathbf{i} + a_2\mathbf{j} \times \mathbf{i} + a_3\mathbf{k} \times \mathbf{i}) \\ &= (-a_2\mathbf{k} - a_3\mathbf{j}) \end{aligned}$$

Thus, $|\mathbf{a} \times \mathbf{i}|^2 = a_2^2 + a_3^2$

Similarly, $|\mathbf{a} \times \mathbf{j}|^2 = a_1^2 + a_3^2$

and $|\mathbf{a} \times \mathbf{k}|^2 = a_1^2 + a_2^2$

Therefore,

$$\begin{aligned} & |\mathbf{a} \times \mathbf{i}|^2 + |\mathbf{a} \times \mathbf{j}|^2 + |\mathbf{a} \times \mathbf{k}|^2 \\ &= (a_2^2 + a_3^2 + a_1^2 + a_3^2 + a_1^2 + a_2^2) \\ &= 2(a_1^2 + a_2^2 + a_3^2) \\ &= 2a^2 \end{aligned}$$

38. Given $\mathbf{a} \cdot \mathbf{b} = 0 = \mathbf{a} \cdot \mathbf{c}$

Thus, $\mathbf{a}$ is perpendicular to $\mathbf{b}$ and $\mathbf{c}$.

A unit vector perpendicular to $\mathbf{b}$ and $\mathbf{c}$

$$\begin{aligned} &= \pm \frac{(\mathbf{b} \times \mathbf{c})}{|\mathbf{b} \times \mathbf{c}|} \\ &= \pm \frac{(\mathbf{b} \times \mathbf{c})}{|\mathbf{b}||\mathbf{c}|\sin\left(\frac{\pi}{6}\right)} \\ &= \pm \frac{(\mathbf{b} \times \mathbf{c})}{1/2} \\ &= \pm 2(\mathbf{b} \times \mathbf{c}). \end{aligned}$$

39. Given,

$$\begin{aligned} & \mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c} \\ \Rightarrow & \mathbf{a} \times (\mathbf{b} - \mathbf{c}) = \mathbf{0} \\ \therefore & (\mathbf{b} - \mathbf{c}) \text{ is parallel to } \mathbf{a} \\ \Rightarrow & (\mathbf{b} - \mathbf{c}) = \lambda \mathbf{a} \\ \Rightarrow & \mathbf{b} - \mathbf{c} + \lambda \mathbf{a}, \lambda \in \mathbb{R} \end{aligned}$$

40. Given,

$$\begin{aligned} & \mathbf{R} \times \mathbf{B} = \mathbf{R} \times \mathbf{C} \\ \Rightarrow & \mathbf{R} \times (\mathbf{B} - \mathbf{C}) = \mathbf{0} \\ \therefore & \mathbf{R} \text{ is parallel to } (\mathbf{B} - \mathbf{C}) \\ \Rightarrow & \mathbf{R} = \lambda(\mathbf{B} - \mathbf{C}) \\ \Rightarrow & \mathbf{R} = \lambda(-3\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}) \end{aligned} \quad \dots (i)$$

Also,

$$\begin{aligned} & \mathbf{R} \cdot \mathbf{A} = 1 \\ \Rightarrow & -6\lambda - 6\lambda = 1 \\ \Rightarrow & \lambda = -\frac{1}{12} \end{aligned}$$

Thus, $\mathbf{R} = -\frac{1}{2}(-3\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}).$

41. Now,

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} \\ &= \mathbf{i} - \mathbf{j} + \mathbf{k} \end{aligned}$$

Unit vector perpendicular to $\mathbf{a}$ and $\mathbf{b}$

$$= \pm \frac{\mathbf{a} \times \mathbf{b}}{|\mathbf{a} \times \mathbf{b}|}$$

$$= \pm (\mathbf{i} - \mathbf{j} + \mathbf{k})$$

Thus, the number of vectors of unit length = 2.

42. As we know that, if $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ form a right handed system, then

$$\mathbf{c} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ 0 & 1 & 0 \end{vmatrix} = -z\mathbf{i} + x\mathbf{k}$$

43. Given,

$$\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a} \text{ and } \mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow \mathbf{r} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b} \text{ and } \mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow \mathbf{r} \times \mathbf{a} = -\mathbf{r} \times \mathbf{b}$$

$$\Rightarrow \mathbf{r} \times \mathbf{a} + \mathbf{r} \times \mathbf{b} = \mathbf{0}$$

$$\Rightarrow \mathbf{r} \times (\mathbf{a} + \mathbf{b}) = \mathbf{0}$$

$$\therefore \mathbf{r} \text{ is parallel to } (\mathbf{a} + \mathbf{b})$$

$$\Rightarrow \mathbf{r} = \lambda(\mathbf{a} + \mathbf{b})$$

$$\Rightarrow \mathbf{r} = \lambda(\mathbf{a} + \mathbf{b})$$

$$\Rightarrow \mathbf{r} \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{b})$$

$$\Rightarrow \mathbf{r} \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b})$$

$$\Rightarrow \mathbf{a} \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b})$$

$$\Rightarrow \lambda = 1$$

Therefore, $\mathbf{r} = (\mathbf{a} \times \mathbf{b})$

$$= 3\mathbf{i} + \mathbf{j} - \mathbf{k}$$

44. Given that

$$\mathbf{r} \times \mathbf{a} = \mathbf{b} \text{ and } \mathbf{r} \cdot \mathbf{a} = 0$$

$$\Rightarrow \mathbf{r} \times \mathbf{a} = \mathbf{b}$$

$$\Rightarrow \mathbf{a} \times (\mathbf{r} \times \mathbf{a}) = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{a})\mathbf{r} - (\mathbf{a} \cdot \mathbf{r})\mathbf{a} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow |\mathbf{a}|^2 \mathbf{r} - 0 = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow \mathbf{r} = \frac{\mathbf{a} \times \mathbf{b}}{|\mathbf{a}|^2}$$

45. Given $\mathbf{a} \times \mathbf{b} = \mathbf{c}$ and $\mathbf{a} \cdot \mathbf{b} = 3$

Let $\mathbf{b} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

Thus, $x + y + z = 3$

and $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix}$

$$\Rightarrow (z - y)\mathbf{i} + (x - z)\mathbf{j} + (y - x)\mathbf{k} = \mathbf{j} - \mathbf{k}$$

Comparing, we get

$$(z - y) = 0, (x - z) = 1, (y - x) = -1$$

Solving, we get

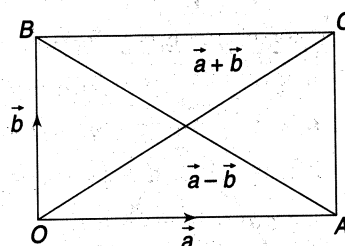
$$\Rightarrow x = \frac{5}{3}, y = \frac{2}{3} = z$$

Thus, $\mathbf{b} = \frac{1}{3}(5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$

46. The vector $[\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}]$ lies in the plane of $\mathbf{a}$ and $\mathbf{b}$ and the vector $(\mathbf{a} \times \mathbf{b})$ is perpendicular to each of $\mathbf{a}$ and $\mathbf{b}$

Thus, the angle between $(\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b})$ and $(\mathbf{a} \times \mathbf{b}) = 90^\circ$.

- 47.



Here, $|\mathbf{a}| = |\mathbf{b}|$ and $(\mathbf{a} \cdot \mathbf{b}) = 0$

Given $\mathbf{a} + \mathbf{b} = 3\mathbf{i} + 4\mathbf{j}$

Now, $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 + 2(\mathbf{a} \cdot \mathbf{b})$

$$\Rightarrow |\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 = 2|\mathbf{a}|^2$$

$$\Rightarrow 2|\mathbf{a}|^2 = |\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a} + \mathbf{b}|^2 = (3^2 + 4^2) = 25$$

$$\Rightarrow |\mathbf{a}|^2 = \frac{25}{2}$$

$$\therefore \text{Area of a square} = |\mathbf{a}|^2 = \frac{25}{2}$$

48. Let $\mathbf{A} = (1, -2, 2)$ and $\mathbf{B} = (2, -1, 3)$

Thus, $\mathbf{r} = \mathbf{BA} = \mathbf{OA} - \mathbf{OB} = -\mathbf{i} - \mathbf{j} - \mathbf{k}$

Therefore, Momentum about the point B

$$= \mathbf{r} \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -1 & -1 \\ 3 & 2 & -4 \end{vmatrix}$$

$$= 6\mathbf{i} - 7\mathbf{j} + \mathbf{k}$$

49. Here, $\mathbf{PQ} = -6\mathbf{i} - \mathbf{j} - \mathbf{k}$

Thus, moment of the couple

$$= \mathbf{M}$$

$$= \mathbf{PQ} \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -6 & -1 & -1 \\ 5 & 0 & 1 \end{vmatrix}$$

$$= -\mathbf{i} + \mathbf{j} + 5\mathbf{k}$$

50. Here, $\mathbf{r} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) - (2\mathbf{i} + \mathbf{j} + \mathbf{k})$

$$\Rightarrow \mathbf{r} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

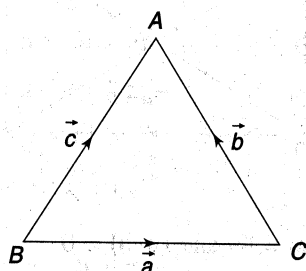
Thus, torque about the point

$$= \mathbf{r} \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 4 & 0 & 1 \end{vmatrix}$$

$$= -2\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}.$$

51. Given $\triangle ABC$, $a = BC$, $b = CA$, $c = AB$



Clearly, $\vec{a} + \vec{b} + \vec{c} = 0$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = |\vec{b} \times \vec{c}| = |\vec{c} \times \vec{a}|$$

$$\Rightarrow |ab \sin(\pi - C)| = |bc \sin(\pi - A)|$$

$$= |ca \sin(\pi - B)|$$

$$\Rightarrow ab \sin C = bc \sin A = ca \sin B$$

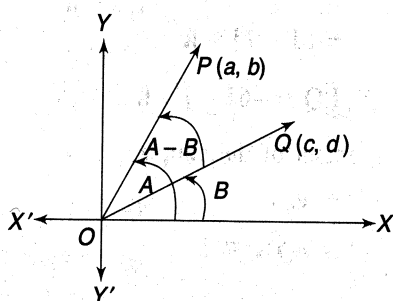
$$\Rightarrow \frac{ab \sin C}{abc} = \frac{bc \sin A}{abc} = \frac{ca \sin B}{abc}$$

$$\Rightarrow \frac{\sin C}{c} = \frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\Rightarrow \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Hence, the result.

52.



Here, $\vec{OP} = a\hat{i} + b\hat{j}$

and $\vec{OQ} = c\hat{i} + d\hat{j}$

Now, $OP \times OQ = (ad - bc)\hat{k}$

$$\Rightarrow [(OP)(OQ)\sin(\angle POQ)]\hat{k} = (ad - bc)\hat{k}$$

$$\Rightarrow -\sin(A - B) = \frac{(ad - bc)}{(OP)(OQ)}$$

$$= \frac{a}{OP} \cdot \frac{d}{OQ} \cdot \frac{b}{OP} \cdot \frac{c}{OQ}$$

$$= \cos A \cdot \sin B - \sin A \cdot \cos B$$

$$\Rightarrow \sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$$

Hence, the result.

54. We have,

$$[\mathbf{a} + \mathbf{b}, \mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}]$$

$$= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})$$

$$= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{c} + \mathbf{c} \times \mathbf{a})$$

$$= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a})$$

$$= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} + \mathbf{a} \cdot \mathbf{b} \times \mathbf{a} + \mathbf{a} \cdot \mathbf{c} \times \mathbf{a})$$

$$+ (\mathbf{b} \cdot \mathbf{b} \times \mathbf{c} + \mathbf{b} \cdot \mathbf{b} \times \mathbf{a} + \mathbf{b} \cdot \mathbf{c} \times \mathbf{a})$$

$$= [\mathbf{a}, \mathbf{b}, \mathbf{c}] + [\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

$$= 2[\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

55. We have,

$$[2\mathbf{a} - \mathbf{b}, 2\mathbf{b} - \mathbf{c}, 2\mathbf{c} - \mathbf{a}]$$

$$= (2\mathbf{a} - \mathbf{b}) \cdot (2\mathbf{b} - \mathbf{c}) \times (2\mathbf{c} - \mathbf{a})$$

$$= (2\mathbf{a} - \mathbf{b}) \cdot (4\mathbf{b} \times \mathbf{c} - 2\mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a})$$

$$= (2\mathbf{a} - \mathbf{b}) \cdot (4\mathbf{b} \times \mathbf{c} + 2\mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a})$$

$$= (8\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} - \mathbf{b} \cdot \mathbf{c} \times \mathbf{a})$$

$$= 7[\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

$$= 0,$$

$\because \mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are coplaner vectors

56. We have,

$$[\mathbf{a}, \mathbf{b} + \mathbf{c}, \mathbf{a} + \mathbf{b} + \mathbf{c}]$$

$$= \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) \times (\mathbf{a} + \mathbf{b} + \mathbf{c})$$

$$= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{c} \times \mathbf{b})$$

$$= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} + \mathbf{a} \cdot \mathbf{c} \times \mathbf{b})$$

$$= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} - \mathbf{a} \cdot \mathbf{c} \times \mathbf{b})$$

$$= 0.$$

57. We have,

$$[\mathbf{a} + \mathbf{b} + \mathbf{c}, \mathbf{a} + \mathbf{b}, \mathbf{a} + \mathbf{c}]$$

$$= (\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot (\mathbf{a} + \mathbf{b}) \times (\mathbf{a} + \mathbf{c})$$

$$= (\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot (\mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c})$$

$$= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} + \mathbf{b} \cdot \mathbf{a} \times \mathbf{c} + \mathbf{c} \cdot \mathbf{b} \times \mathbf{a})$$

$$\begin{aligned}
 &= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} - \mathbf{a} \cdot \mathbf{b} \times \mathbf{c} + \mathbf{c} \cdot \mathbf{b} \times \mathbf{a}) \\
 &= [\mathbf{a}, \mathbf{b}, \mathbf{c}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}] + [\mathbf{a}, \mathbf{b}, \mathbf{c}] \\
 &= [\mathbf{a}, \mathbf{b}, \mathbf{c}]
 \end{aligned}$$

58. We have,

$$\begin{aligned}
 &[\mathbf{a} - \mathbf{b}, \mathbf{b} - \mathbf{c}, \mathbf{c} - \mathbf{a}] \\
 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{b} - \mathbf{c}) \times (\mathbf{c} - \mathbf{a}) \\
 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a}) \\
 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a}) \\
 &= (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} - \mathbf{b} \cdot \mathbf{c} \times \mathbf{a}) \\
 &= [\mathbf{a}, \mathbf{b}, \mathbf{c}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}] \\
 &= 0.
 \end{aligned}$$

59. We have,

$$\begin{aligned}
 &\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = |\mathbf{a}| |\mathbf{b}| |\mathbf{c}| \\
 \Rightarrow &(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = |\mathbf{a}| |\mathbf{b}| |\mathbf{c}| \\
 \Rightarrow &|\mathbf{a}| |\mathbf{b}| |\mathbf{c}| \sin(\theta) \cos(\phi) = |\mathbf{a}| |\mathbf{b}| |\mathbf{c}| \\
 \Rightarrow &\sin(\theta) \cos(\phi) = 1 \\
 \Rightarrow &\sin(\theta) = 1, \cos(\phi) = 1 \\
 \Rightarrow &\theta = \frac{\pi}{2}, \phi = 0
 \end{aligned}$$

Hence, the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\theta = \frac{\pi}{2}$.

60. We have,

$$\begin{aligned}
 &\frac{1}{2}(\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c}) \\
 &= \frac{1}{2}((\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} + (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{b} + (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{c}) \\
 &= \frac{1}{2}((\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} + 0 + 0) \\
 &= \frac{1}{2}(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} \\
 &= \frac{1}{2}[\mathbf{a}, \mathbf{b}, \mathbf{c}] \\
 &= \frac{1}{2} \times 4 = 2
 \end{aligned}$$

61. Since the given vectors are coplanar, so

$$\begin{aligned}
 &\begin{vmatrix} 2 & -1 & \lambda \\ 1 & -1 & 2 \\ 3 & -2 & 1 \end{vmatrix} = 0 \\
 \Rightarrow &6 - 5 + \lambda(-2 + 3) = 0 \\
 \Rightarrow &1 + \lambda = 0 \\
 \Rightarrow &\lambda = -1
 \end{aligned}$$

62. The volume of the parallelopiped

$$\begin{aligned}
 &= [\mathbf{a}, \mathbf{b}, \mathbf{c}] \\
 &= \begin{vmatrix} 3 & -2 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -2 \end{vmatrix}
 \end{aligned}$$

$$\begin{aligned}
 &= 3(-3) + 2(-3) + 0 \\
 &= -3 = 3\text{c.u.}
 \end{aligned}$$

63. We have,

$$\begin{aligned}
 &2(\mathbf{a} + \mathbf{b} + \mathbf{c}) = 6\mathbf{i} + 8\mathbf{j} - 12\mathbf{k} \\
 \Rightarrow &(\mathbf{a} + \mathbf{b} + \mathbf{c}) = 3\mathbf{i} + 4\mathbf{j} - 6\mathbf{k} \\
 \text{Thus, } &\mathbf{a} = \mathbf{i} + \mathbf{j} - \mathbf{k}, \mathbf{b} = \mathbf{k}, \mathbf{c} = 2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k} \\
 \text{Hence, the volume of the parallelopiped}
 \end{aligned}$$

$$= \begin{vmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 2 & 3 & -5 \end{vmatrix} = 1 \text{ c.u.}$$

64. Given $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = 5$

$$\begin{aligned}
 \text{Now, } &[3(\mathbf{a} + \mathbf{b}), (\mathbf{b} + \mathbf{c}), 2(\mathbf{c} + \mathbf{a})] \\
 &= 3 \cdot 2 [(\mathbf{a} + \mathbf{b}), (\mathbf{b} + \mathbf{c}), 2(\mathbf{c} + \mathbf{a})] \\
 &= 6 [(\mathbf{a} + \mathbf{b}), (\mathbf{b} + \mathbf{c}), 2(\mathbf{c} + \mathbf{a})] \\
 &= 6 \times 2 [\mathbf{a}, \mathbf{b}, \mathbf{c}] \\
 &= 6 \times 2 \times 5 \\
 &= 60.
 \end{aligned}$$

65. The volume of the parallelopiped,

$$\begin{aligned}
 V &= \begin{vmatrix} 1 & a & 1 \\ 0 & 1 & a \\ a & 0 & 1 \end{vmatrix} \\
 &= 1 + a(a^2 - 1) \\
 &= a^3 - a + 1
 \end{aligned}$$

$$\Rightarrow \frac{dV}{da} = 3a^2 - 1$$

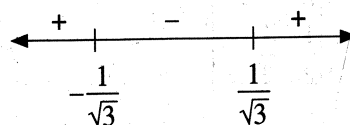
For maximum or minimum,

$$\frac{dV}{da} = 0$$

gives

$$\Rightarrow 3a^2 - 1 = 0$$

$$\Rightarrow a = \pm \frac{1}{\sqrt{3}}$$



By sign scheme, the point of minima is $a = \frac{1}{\sqrt{3}}$.

67. Since the given vectors are coplanar, so

$$\begin{aligned}
 &\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \\
 \Rightarrow &\begin{vmatrix} a & 1-a & 1-a \\ 1 & b-1 & 0 \\ 1 & 0 & c-1 \end{vmatrix} = 0, \begin{pmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{pmatrix}
 \end{aligned}$$

$$\Rightarrow a(b-1)(c-1)(a-1)(c-1) + (1-b)(1-b) = 0.$$

$$\Rightarrow (1-b)(1-c) + (1-a)(1-c) + (1-a)(1-b) = 0$$

Dividing both the sides by $(1-a)(1-b)(1-c)$, we get

$$\frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$$

$$\Rightarrow \left(1 + \frac{a}{1-a}\right) + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

$$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

68. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$,

$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$

We have, $[\mathbf{a}, \mathbf{b}, \mathbf{c}]^2$

$$= [\mathbf{a}, \mathbf{b}, \mathbf{c}][\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \times \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\begin{vmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{vmatrix}$$

$$= \begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix}$$

69. As we know that,

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}]^2 = \begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= \left(1 - \frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{4} - \frac{1}{2}\right)$$

$$= \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$

$$\text{Thus, } [\mathbf{a}, \mathbf{b}, \mathbf{c}] = \frac{\sqrt{3}}{2\sqrt{2}}$$

Therefore, the volume of the parallelepiped

$$= \frac{\sqrt{3}}{2\sqrt{2}}$$

70. Let $\mathbf{a} \times \mathbf{b} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

We have,

$$[\mathbf{a}, \mathbf{b}, \mathbf{i}] = ((\mathbf{a} \times \mathbf{b}) \cdot \mathbf{i})$$

$$= a_1$$

Similarly, $[\mathbf{a}, \mathbf{b}, \mathbf{j}] = a_2$

and $[\mathbf{a}, \mathbf{b}, \mathbf{k}] = a_3$

Thus,

$$[\mathbf{a}, \mathbf{b}, \mathbf{i}]\mathbf{i} + [\mathbf{a}, \mathbf{b}, \mathbf{j}]\mathbf{j} + [\mathbf{a}, \mathbf{b}, \mathbf{k}]\mathbf{k}$$

$$= a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$$

$$= (\mathbf{a} \times \mathbf{b})$$

Hence, the result.

71. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$,

$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$

Also, let $\mathbf{m} = m_1\mathbf{i} + m_2\mathbf{j} + m_3\mathbf{k}$

and $\mathbf{n} = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$

Now,

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}](\mathbf{m} \times \mathbf{n})$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix}$$

$$= \begin{vmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \\ \mathbf{a} \cdot \mathbf{m} & \mathbf{b} \cdot \mathbf{m} & \mathbf{c} \cdot \mathbf{m} \\ \mathbf{a} \cdot \mathbf{n} & \mathbf{b} \cdot \mathbf{n} & \mathbf{c} \cdot \mathbf{n} \end{vmatrix}$$

72. Given $\mathbf{r} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$

Taking scalar product of $(\mathbf{b} \times \mathbf{c})$

$$\mathbf{r} \cdot (\mathbf{b} \times \mathbf{c}) = x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + y\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) + z\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c})$$

$$\Rightarrow \mathbf{r} \cdot (\mathbf{b} \times \mathbf{c}) = x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$$

$$\Rightarrow [\mathbf{r}, \mathbf{b}, \mathbf{c}] = x[\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

$$\Rightarrow x = \frac{[\mathbf{r}, \mathbf{b}, \mathbf{c}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$$

$$\text{Similarly, } y = \frac{[\mathbf{r}, \mathbf{c}, \mathbf{a}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} \text{ \& } z = \frac{[\mathbf{r}, \mathbf{a}, \mathbf{b}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$$

$$\text{Hence, } \mathbf{r} = \frac{[\mathbf{r}, \mathbf{b}, \mathbf{c}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} \mathbf{a} + \frac{[\mathbf{r}, \mathbf{c}, \mathbf{a}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} \mathbf{b} + \frac{[\mathbf{r}, \mathbf{a}, \mathbf{b}]}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} \mathbf{c}$$

73. Given,

$$[\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b}] = \frac{1}{4}$$

$$\Rightarrow (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{b}) = \frac{1}{4}$$

$$\Rightarrow |(\mathbf{a} \times \mathbf{b})|^2 = \frac{1}{4}$$

$$\Rightarrow |(\mathbf{a} \times \mathbf{b})| = \frac{1}{2}$$

$$\Rightarrow ab \sin(\theta) = \frac{1}{2}$$

$$\Rightarrow \sin(\theta) = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

74. Given $\alpha = \mathbf{x}(\mathbf{a} \times \mathbf{b}) + \mathbf{y}(\mathbf{b} \times \mathbf{c}) + \mathbf{z}(\mathbf{c} \times \mathbf{a})$

$$\Rightarrow \alpha \cdot \mathbf{c} = \mathbf{x}(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} + \mathbf{y}(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{c} + \mathbf{z}(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{c}$$

$$= \mathbf{x}(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{x}[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$$

$$\Rightarrow x = \frac{\alpha \cdot \mathbf{c}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$$

Similarly, $y = \frac{\alpha \cdot \mathbf{c}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$, $z = \frac{\alpha \cdot \mathbf{b}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$

Thus,

$$x + y + z$$

$$= \frac{\alpha \cdot \mathbf{c}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} + \frac{\alpha \cdot \mathbf{a}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]} + \frac{\alpha \cdot \mathbf{b}}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$$

$$= \frac{\alpha \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c})}{[\mathbf{a}, \mathbf{b}, \mathbf{c}]}$$

$$= \frac{\alpha \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c})}{1/8}$$

$$= 8\alpha \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c}).$$

75. Volume of the tetrahedron

$$= \frac{1}{6} [\mathbf{a}, \mathbf{b}, \mathbf{c}]$$

$$= \frac{1}{6} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= \frac{1}{6}(-1 + 2 + 3)$$

$$= \frac{2}{3} \text{ c.u.}$$

76. Given $|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 = 9$

$$\Rightarrow 2(a^2 + b^2 + c^2) - 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 9$$

$$\Rightarrow 6 - 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 9$$

$$\Rightarrow 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = -3$$

$$\Rightarrow (a^2 + b^2 + c^2) + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = 3 - 3$$

$$\Rightarrow |\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = 0$$

$$\Rightarrow (\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{0}$$

Now,

$$|2\mathbf{a} + 5\mathbf{b} + 5\mathbf{c}| = |2\mathbf{a} + 5(\mathbf{b} + \mathbf{c})|$$

$$= |2\mathbf{a} + 5(-\mathbf{a})|$$

$$= |-3(\mathbf{a})|$$

$$= |-3||\mathbf{a}|$$

$$= 3.$$

77. Let $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = \mathbf{k} + \mathbf{i}$

Now,

$$(\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = (\mathbf{i} - \mathbf{j} + \mathbf{k})$$

$$\hat{\mathbf{m}} = \frac{(\mathbf{a} \times \mathbf{b})}{|(\mathbf{a} \times \mathbf{b})|} = \frac{1}{\sqrt{3}}(\mathbf{i} - \mathbf{j} + \mathbf{k})$$

similarly, $\hat{\mathbf{n}} = \frac{(\mathbf{b} \times \mathbf{c})}{|(\mathbf{b} \times \mathbf{c})|} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} - \mathbf{k})$

and $\hat{\mathbf{p}} = \frac{(\mathbf{c} \times \mathbf{a})}{|(\mathbf{c} \times \mathbf{a})|} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} - \mathbf{k})$

Volume of a parallelepiped

$$= \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$$

$$= \frac{1}{3\sqrt{3}} \begin{vmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 1 \end{vmatrix} (C_2 \rightarrow C_2 + C_1)$$

$$= \frac{2}{3\sqrt{3}} (1 + 1)$$

$$= \frac{4}{3\sqrt{3}} \text{ c.u.}$$

78. Given $\mathbf{a} + \mathbf{b} + \mathbf{c} = \alpha \mathbf{d}$

and $\mathbf{b} + \mathbf{c} + \mathbf{d} = \beta \mathbf{a}$

$$\Rightarrow \mathbf{d} = -\mathbf{b} - \mathbf{c} - \beta \mathbf{a}$$

$$\Rightarrow \alpha \mathbf{d} = -\alpha \mathbf{b} - \alpha \mathbf{c} - \alpha \beta \mathbf{a}$$

$$\Rightarrow \mathbf{a} + \mathbf{b} + \mathbf{c} = \alpha \mathbf{b} - \alpha \mathbf{c} - \alpha \beta \mathbf{a}$$

Comparing, the co-efficients of the vectors, we get

$$\alpha = -1, \alpha \beta = -1$$

$$\Rightarrow \alpha = -1, \beta = 1$$

Now,

$$\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} + 4 = \alpha \mathbf{d} + \mathbf{d} + 4$$

$$= -\mathbf{d} + \mathbf{d} + 4$$

$$= 4.$$

79. Let $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$,

$$\mathbf{b} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$$

and $\mathbf{c} = \mathbf{i} - \mathbf{j} + \mathbf{k}$.

We have to find $\pm \frac{\mathbf{a} \times (\mathbf{b} \times \mathbf{c})}{|\mathbf{a} \times (\mathbf{b} \times \mathbf{c})|}$.

Now,

$$\begin{aligned} \mathbf{b} \times \mathbf{c} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} \\ &= 2\mathbf{i} - \mathbf{j} - 3\mathbf{k} \end{aligned}$$

Also,

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 2 & 6 \\ 2 & -1 & -3 \end{vmatrix} \\ &= 21\mathbf{i} - 7\mathbf{k} \end{aligned}$$

Thus,

$$\begin{aligned} &\pm \frac{\mathbf{a} \times (\mathbf{b} \times \mathbf{c})}{|\mathbf{a} \times (\mathbf{b} \times \mathbf{c})|} \\ &= \pm \frac{(21\mathbf{i} - 7\mathbf{k})}{\sqrt{441 + 49}} \\ &= \pm \frac{(21\mathbf{i} - 7\mathbf{k})}{7\sqrt{10}} \\ &= \pm \frac{(3\mathbf{i} - \mathbf{k})}{\sqrt{10}} \end{aligned}$$

80. We have,

$$\begin{aligned} \mathbf{a} \times (\mathbf{a} \times \mathbf{b}) &= (\mathbf{a} \cdot \mathbf{b})\mathbf{a} - (\mathbf{a} \cdot \mathbf{a})\mathbf{b} \\ &= \mathbf{a} - |\mathbf{a}|^2\mathbf{b} \\ &= \mathbf{a} - 3\mathbf{b} \end{aligned}$$

$$\Rightarrow 3\mathbf{b} = \mathbf{a} - \mathbf{a} \times (\mathbf{a} \times \mathbf{b}) \quad \dots(i)$$

Now,

$$\begin{aligned} \mathbf{a} \times (\mathbf{a} \times \mathbf{b}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} \\ &= -2\mathbf{i} + \mathbf{j} + \mathbf{k} \end{aligned}$$

Thus, from Eq. (i), we get

$$3\mathbf{b} = (\mathbf{i} + \mathbf{j} + \mathbf{k}) - (-2\mathbf{i} + \mathbf{j} + \mathbf{k})$$

$$\Rightarrow 3\mathbf{b} = 3\mathbf{i}$$

$$\Rightarrow \mathbf{b} = \mathbf{i}$$

81. We have,

$$(\mathbf{a} \times (\mathbf{b} \times \mathbf{c})) = \frac{1}{2}\mathbf{b}$$

$$\begin{aligned} \Rightarrow (\mathbf{a} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} &= \frac{1}{2}\mathbf{b} \\ &= \frac{1}{2}, (\mathbf{a} \cdot \mathbf{b}) = 0 \end{aligned}$$

$$\Rightarrow a c \cos(\theta_1) = \frac{1}{2} \text{ and } a b \cos(\theta_2) = 0$$

$$\Rightarrow \cos(\theta_1) = \frac{1}{2} \text{ and } \cos(\theta_2) = 0$$

$$\Rightarrow (\theta_1) = \frac{\pi}{3}, (\theta_2) = \frac{\pi}{2}$$

Hence, the result.

82. We have,

$$(\mathbf{a} \times (\mathbf{b} \times \mathbf{c})) = \frac{1}{\sqrt{2}}(\mathbf{b} + \mathbf{c})$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = \frac{1}{\sqrt{2}}(\mathbf{b} + \mathbf{c})$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{c}) = \frac{1}{\sqrt{2}}, (\mathbf{a} \cdot \mathbf{b}) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow a c \cos(\theta_1) = \frac{1}{\sqrt{2}} \text{ and } a b \cos(\theta_2) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos(\theta_1) = \frac{1}{\sqrt{2}} \text{ and } \cos(\theta_2) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta_1 = \frac{\pi}{4} \text{ and } \theta_2 = \frac{\pi}{4}$$

$$\Rightarrow \theta_2 = \frac{\pi}{4}$$

Hence, the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{4}$

83. Given,

$$\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) + \mathbf{b} = \mathbf{0}$$

$$\Rightarrow \mathbf{a} \times (\mathbf{a} \times \mathbf{c}) = -\mathbf{b}$$

$$\Rightarrow |\mathbf{a} \times (\mathbf{a} \times \mathbf{c})| = |\mathbf{b}|$$

$$\Rightarrow |\mathbf{a}|(|\mathbf{a} \times \mathbf{c}|) \sin\left(\frac{\pi}{2}\right) = |\mathbf{b}|$$

$$\Rightarrow |\mathbf{a}|(|\mathbf{a} \times \mathbf{c}|) = |\mathbf{b}|$$

$$\Rightarrow |\mathbf{a} \times \mathbf{c}| = 1$$

$$\Rightarrow a c \sin \theta = 1$$

$$\Rightarrow \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

Hence, the angle between $\mathbf{a}$ and $\mathbf{c}$ is $\frac{\pi}{6}$.

84. We have,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = 0$$

$$\Rightarrow \{(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}\} \cdot \{(\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{c} \cdot \mathbf{b})\mathbf{a}\} = 0$$

$$\Rightarrow ((\mathbf{a} \cdot \mathbf{c})\mathbf{b})^2 - (\mathbf{a} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{a})$$

$$- (\mathbf{a} \cdot \mathbf{b})(\mathbf{c} \cdot \mathbf{a})(\mathbf{c} \cdot \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{c} \cdot \mathbf{b})(\mathbf{c} \cdot \mathbf{a}) = 0$$

$$\Rightarrow ((\mathbf{a} \cdot \mathbf{c})\mathbf{b})^2 - (\mathbf{a} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{a}) = 0$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{c})\{(\mathbf{a} \cdot \mathbf{c})\mathbf{b}^2 - (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})\} = 0$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{c}) = 0, \{(\mathbf{a} \cdot \mathbf{c})\mathbf{b}^2 - (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})\} = 0$$

Hence, $(\mathbf{a} \cdot \mathbf{c}) = 0$

85. Given,

$$\begin{aligned}
 (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} &= \frac{1}{3} \|\mathbf{b}\| \|\mathbf{c}\| \mathbf{a} \\
 \Rightarrow \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) &= -\frac{1}{3} \|\mathbf{b}\| \|\mathbf{c}\| \mathbf{a} \\
 \Rightarrow (\mathbf{c} \cdot \mathbf{b}) \mathbf{a} - (\mathbf{a} \cdot \mathbf{a}) \mathbf{b} &= -\frac{1}{3} \|\mathbf{b}\| \|\mathbf{c}\| \mathbf{a} \\
 \Rightarrow (\mathbf{c} \cdot \mathbf{b}) &= -\frac{1}{3} \|\mathbf{b}\| \|\mathbf{c}\| \\
 \Rightarrow \|\mathbf{c}\| \|\mathbf{b}\| \cos(\theta) &= -\frac{1}{3} \|\mathbf{b}\| \|\mathbf{c}\| \\
 \Rightarrow \cos \theta &= -\frac{1}{3} \\
 \Rightarrow \sin \theta &= \sqrt{1 - \frac{1}{9}} = \frac{2\sqrt{2}}{3}
 \end{aligned}$$

Hence, the result.

86. Given,

$$\begin{aligned}
 \mathbf{a} \cdot \mathbf{b} &= 0 \\
 \text{Let } \mathbf{r} &= m\mathbf{a} + n\mathbf{b} + p(\mathbf{a} \times \mathbf{b}) \quad \dots(i) \\
 \text{Now, } \mathbf{r} \cdot \mathbf{a} &= m(\mathbf{a} \cdot \mathbf{a}) + n(\mathbf{a} \cdot \mathbf{b}) + p(\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b})) \\
 \Rightarrow m &= 0 \\
 \text{Also, } \mathbf{r} \cdot \mathbf{b} &= m(\mathbf{a} \cdot \mathbf{b}) + n(\mathbf{b} \cdot \mathbf{b}) + p(\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b})) \\
 \Rightarrow 1 &= 0 + n(\mathbf{b} \cdot \mathbf{b}) + 0 \\
 \Rightarrow n &= 1 \\
 \text{Again, } \mathbf{r} \cdot \mathbf{a} \times \mathbf{b} &= m(\mathbf{a} \cdot \mathbf{a} \times \mathbf{b}) + n(\mathbf{b} \cdot \mathbf{a} \times \mathbf{b}) \\
 &\quad + p[(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{b})] \\
 \Rightarrow 1 &= 0 + 0 + p\|\mathbf{a} \times \mathbf{b}\|^2 \\
 \Rightarrow p &= 1
 \end{aligned}$$

Hence, from Eq. (i),

$$\mathbf{r} = \mathbf{b} + (\mathbf{a} \times \mathbf{b})$$

87. We have,

$$\begin{aligned}
 \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) &= \frac{1}{2} \mathbf{a} \\
 \Rightarrow (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} + (\mathbf{b} \cdot \mathbf{a}) \mathbf{c} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{a} &= \frac{1}{2} \mathbf{a} \\
 \Rightarrow (\mathbf{a} \cdot \mathbf{c}) = 0, (\mathbf{a} \cdot \mathbf{b}) = 0, (\mathbf{b} \cdot \mathbf{a}) = 0, (\mathbf{b} \cdot \mathbf{c}) &= -\frac{1}{2}
 \end{aligned}$$

Now, $(\mathbf{a} \cdot \mathbf{c}) = 0$

$$\Rightarrow a c \cos(\theta_1) = 0$$

$$\Rightarrow \cos(\theta_1) = 0$$

$$= \cos\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \theta_1 = \left(\frac{\pi}{2}\right)$$

Thus,

$$\begin{aligned}
 |(\mathbf{a} \times \mathbf{c})| &= a c \sin\left(\frac{\pi}{2}\right) \\
 &= 7 \cdot 1 \cdot 1 \\
 &= 7.
 \end{aligned}$$

88. We have,

$$\begin{aligned}
 (\mathbf{d} + \mathbf{a}) \cdot [\mathbf{a} \times \mathbf{b} \times (\mathbf{c} \times \mathbf{d})] &= (\mathbf{d} + \mathbf{a}) \cdot (\mathbf{a} \times [(\mathbf{b} \cdot \mathbf{d}) \mathbf{c} - (\mathbf{b} \times \mathbf{c}) \mathbf{d}]) \\
 &= (\mathbf{d} + \mathbf{a}) \cdot [(\mathbf{a} \times \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \times \mathbf{d})(\mathbf{b} \cdot \mathbf{c})] \\
 &= [\mathbf{d} \cdot (\mathbf{a} \times \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) + \mathbf{a} \cdot (\mathbf{a} \times \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) \\
 &\quad - \mathbf{d} \cdot (\mathbf{a} \times \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) - \mathbf{a} \cdot (\mathbf{a} \times \mathbf{d})(\mathbf{b} \cdot \mathbf{c})] \\
 &= [\mathbf{a}, \mathbf{c}, \mathbf{d}](\mathbf{b} \cdot \mathbf{d})
 \end{aligned}$$

89. Given,

$$\begin{aligned}
 p\mathbf{x} + (\mathbf{x} \times \mathbf{a}) &= \mathbf{b} \\
 \Rightarrow p(\mathbf{a} \cdot \mathbf{x}) + \mathbf{a} \cdot (\mathbf{x} \times \mathbf{a}) &= \mathbf{a} \cdot \mathbf{b} \\
 \Rightarrow p(\mathbf{a} \cdot \mathbf{x}) + 0 &= \mathbf{a} \cdot \mathbf{b} \\
 \Rightarrow p(\mathbf{a} \cdot \mathbf{x}) &= \mathbf{a} \cdot \mathbf{b} \\
 \Rightarrow p &= \frac{(\mathbf{a} \cdot \mathbf{b})}{(\mathbf{a} \cdot \mathbf{x})} \quad \dots(i) \\
 \text{Also, } p\mathbf{x} + (\mathbf{x} \times \mathbf{a}) &= \mathbf{b} \\
 \Rightarrow p(\mathbf{a} \times \mathbf{x}) + \mathbf{a} \times (\mathbf{x} \times \mathbf{a}) &= \mathbf{a} \times \mathbf{b} \\
 \Rightarrow p(\mathbf{a} \times \mathbf{x}) + (\mathbf{a} \cdot \mathbf{a}) \mathbf{x} - (\mathbf{a} \cdot \mathbf{x}) \mathbf{a} &= \mathbf{a} \times \mathbf{b} \\
 \Rightarrow (\mathbf{a} \cdot \mathbf{a}) \mathbf{x} = (\mathbf{a} \cdot \mathbf{x}) \mathbf{a} - \mathbf{a} \times \mathbf{b} - p(\mathbf{a} \times \mathbf{x}) \\
 \Rightarrow (\mathbf{a} \cdot \mathbf{a}) \mathbf{x} &= \frac{(\mathbf{a} \cdot \mathbf{b})}{p} \mathbf{a} - \mathbf{a} \times \mathbf{b} - p(\mathbf{a} \times \mathbf{x}) \\
 \Rightarrow (\mathbf{a} \cdot \mathbf{a}) \mathbf{x} &= \frac{(\mathbf{a} \cdot \mathbf{b})}{p} \mathbf{a} - \mathbf{a} \times \mathbf{b} - p(p\mathbf{x} - \mathbf{b}) \\
 \Rightarrow (a^2 + p^2) \mathbf{x} &= \frac{(\mathbf{a} \cdot \mathbf{b})}{p} \mathbf{a} - \mathbf{a} \times \mathbf{b} - p\mathbf{b} \\
 \Rightarrow \mathbf{x} &= \frac{\frac{(\mathbf{a} \cdot \mathbf{b})}{p} \mathbf{a} - \mathbf{a} \times \mathbf{b} - p\mathbf{b}}{(a^2 + p^2)}
 \end{aligned}$$

90. Given $[\mathbf{x} \ \mathbf{a} \ \mathbf{b}] = 0$ $\Rightarrow \mathbf{x}, \mathbf{a}, \mathbf{b}$ are coplanar vectors

$$\Rightarrow \mathbf{x} = m\mathbf{a} + n\mathbf{b}$$

$$\Rightarrow \mathbf{x} \cdot \mathbf{a} = m(\mathbf{a} \cdot \mathbf{a}) + n(\mathbf{b} \cdot \mathbf{a})$$

$$\Rightarrow \mathbf{x} \cdot \mathbf{a} = m(\mathbf{a} \cdot \mathbf{a}) + n(\mathbf{b} \cdot \mathbf{a})$$

$$\Rightarrow 0 = m\|\mathbf{a}\|^2 + 0$$

$$\Rightarrow m = 0$$

Also, $\mathbf{x} = m\mathbf{a} + n\mathbf{b}$

$$\Rightarrow \mathbf{x} \cdot \mathbf{b} = m(\mathbf{a} \cdot \mathbf{b}) + n(\mathbf{b} \cdot \mathbf{b})$$

$$\Rightarrow 1 = 0 + |b|^2$$

$$\Rightarrow n = \frac{1}{|b|^2}$$

Thus, $\mathbf{x} = m\mathbf{a} + n\mathbf{b} = \frac{\mathbf{b}}{|b|^2} = \frac{\hat{\mathbf{b}}}{|b|}$

91. We have,

$$[\mathbf{a} \times (\mathbf{b} + \mathbf{c}), \mathbf{b} \times (\mathbf{c} - 2\mathbf{a}), \mathbf{c} \times (\mathbf{a} + 3\mathbf{b})]$$

$$= [\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}, \mathbf{b} \times \mathbf{c} - 2\mathbf{b} \times \mathbf{a}, \mathbf{c} \times \mathbf{a} + 3\mathbf{c} \times \mathbf{b}]$$

$$= [\mathbf{a} \times \mathbf{b} - \mathbf{c} \times \mathbf{a}, \mathbf{b} \times \mathbf{c} + 2\mathbf{a} \times \mathbf{b}, \mathbf{c} \times \mathbf{a} - 3\mathbf{b} \times \mathbf{c}]$$

$$= [\mathbf{x} - \mathbf{z}, \mathbf{y} + 2\mathbf{x}, \mathbf{z} - 3\mathbf{y}]$$

$$\text{where } \mathbf{x} = \mathbf{a} \times \mathbf{b}, \mathbf{y} = \mathbf{b} \times \mathbf{c}, \mathbf{z} = \mathbf{c} \times \mathbf{a}$$

$$= (\mathbf{x} - \mathbf{z}) \cdot (\mathbf{y} + 2\mathbf{x}) \times (\mathbf{z} - 3\mathbf{y})$$

$$= (\mathbf{x} - \mathbf{z}) \cdot (\mathbf{y} \times \mathbf{z} - 6\mathbf{x} \times \mathbf{y} + 2\mathbf{x} \times \mathbf{z})$$

$$= \mathbf{x} \cdot (\mathbf{y} \times \mathbf{z}) + 6\mathbf{z} \cdot (\mathbf{x} \times \mathbf{y})$$

$$= 7\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z})$$

$$= 7[\mathbf{x} \cdot \mathbf{y} \times \mathbf{z}]$$

$$= 7[\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}]^2$$

$$= 28$$

92. We have,

$$[(\mathbf{a} - \mathbf{b}), (\mathbf{a} - \mathbf{b} - \mathbf{c}), (\mathbf{a} + 2\mathbf{b} - \mathbf{c})]$$

$$= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b} - \mathbf{c}) \times (\mathbf{a} + 2\mathbf{b} - \mathbf{c})$$

$$= (\mathbf{a} - \mathbf{b}) \cdot [2\mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c} - \mathbf{c} \times \mathbf{a} - 2\mathbf{c} \times \mathbf{b}]$$

$$= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) - 2\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b}) + \mathbf{b} \cdot (\mathbf{a} \times \mathbf{c}) + \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$$

$$= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + 2\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) - \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) + \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$$

$$= 3[\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}]$$

$$= 15$$

93. We have,

$$[(2\mathbf{a} \times 3\mathbf{b}), (3\mathbf{b} \times 4\mathbf{c}), (4\mathbf{c} \times 2\mathbf{a})]$$

$$= [\mathbf{x} \times \mathbf{y}, \mathbf{y} \times \mathbf{z}, \mathbf{z} \times \mathbf{x}]$$

$$\text{where } 2\mathbf{a} = \mathbf{x}, 3\mathbf{b} = \mathbf{y}, 4\mathbf{c} = \mathbf{z}$$

$$= [\mathbf{x}, \mathbf{y}, \mathbf{z}]^2$$

$$= [2\mathbf{a}, 3\mathbf{b}, 4\mathbf{c}]^2$$

$$= 4 \cdot 9 \cdot 16[\mathbf{a}, \mathbf{b}, \mathbf{c}]^2$$

$$= 576 \times 4 = 2304$$

94. Given,

$$\mathbf{x} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{x} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$$

$$\Rightarrow (\mathbf{x} - \mathbf{a}) = \lambda \mathbf{b}, \quad \text{where } \lambda \in \mathbb{R}$$

$$\Rightarrow \mathbf{x} = \mathbf{a} + \lambda \mathbf{b}$$

95. We have,

$$\mathbf{x} \times \mathbf{a} = \mathbf{b}$$

$$\Rightarrow (\mathbf{x} \times \mathbf{a}) \times \mathbf{a} = \mathbf{b} \times \mathbf{a}$$

$$\Rightarrow \mathbf{a} \times (\mathbf{x} \times \mathbf{a}) = -\mathbf{b} \times \mathbf{a} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{a})\mathbf{x} - (\mathbf{a} \cdot \mathbf{x})\mathbf{a} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow |\mathbf{a}|^2 \mathbf{x} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow \mathbf{x} = \frac{\mathbf{a} \times \mathbf{b}}{|\mathbf{a}|^2}$$

96. Given,

$$\mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$$

$$\Rightarrow (\mathbf{r} - \mathbf{a}) = \lambda \mathbf{b}, \lambda \in \mathbb{R}$$

$$\Rightarrow \mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$$

$$\Rightarrow \mathbf{r} \cdot \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) + \lambda(\mathbf{b} \cdot \mathbf{c})$$

$$\Rightarrow \lambda = -\frac{(\mathbf{a} \cdot \mathbf{c})}{(\mathbf{b} \cdot \mathbf{c})}$$

$$\text{Thus, } \mathbf{r} = \mathbf{a} - \left(\frac{(\mathbf{a} \cdot \mathbf{c})}{(\mathbf{b} \cdot \mathbf{c})} \right) \mathbf{b}$$

97. Given,

$$x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = \mathbf{d}$$

Taking scalar product of $(\mathbf{b} \times \mathbf{c})$, we get

$$x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + y\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) + z\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{d} \cdot (\mathbf{b} \times \mathbf{c})$$

$$\Rightarrow x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + 0 + 0 = \mathbf{d} \cdot (\mathbf{b} \times \mathbf{c})$$

$$\Rightarrow x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{d} \cdot (\mathbf{b} \times \mathbf{c})$$

$$\Rightarrow x = \frac{\mathbf{d} \cdot (\mathbf{b} \times \mathbf{c})}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})} = \frac{[\mathbf{d} \mathbf{b} \mathbf{c}]}{[\mathbf{a} \mathbf{b} \mathbf{c}]}$$

Similarly, we can easily find that

$$y = \frac{[\mathbf{d} \mathbf{c} \mathbf{a}]}{[\mathbf{a} \mathbf{b} \mathbf{c}]} \text{ and } z = \frac{[\mathbf{d} \mathbf{a} \mathbf{b}]}{[\mathbf{a} \mathbf{b} \mathbf{c}]}$$

98. We have,

$$\mathbf{x} + \mathbf{y} = \mathbf{a}$$

$$\Rightarrow (\mathbf{x} + \mathbf{y}) \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{a}$$

$$\Rightarrow \mathbf{x} \cdot \mathbf{a} + \mathbf{y} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{a}$$

$$\Rightarrow 1 + \mathbf{y} \cdot \mathbf{a} = |\mathbf{a}|^2 \quad \dots(i)$$

$$\text{Also, } \mathbf{x} \times \mathbf{y} = \mathbf{b}$$

$$\Rightarrow \mathbf{a} \times (\mathbf{x} \times \mathbf{y}) = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (\mathbf{a} \cdot \mathbf{y})\mathbf{x} - (\mathbf{a} \cdot \mathbf{x})\mathbf{y} = \mathbf{a} \times \mathbf{b}$$

$$\Rightarrow (|\mathbf{a}|^2 - 1)\mathbf{x} - \mathbf{y} = \mathbf{a} \times \mathbf{b} \text{ [from Eq. (i)] } \dots(ii)$$

$$\text{Again } \mathbf{x} + \mathbf{y} = \mathbf{a} \quad \dots(iii)$$

Adding Eqs. (ii) and (iii), we get

$$(|\mathbf{a}|^2)\mathbf{x} = (\mathbf{a} \times \mathbf{b} + \mathbf{a})$$